[INCH-POUND]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

MILLING MACHINE, VERTICAL KNEE TYPE, 3 AXIS, COMPUTER NUMERICAL CONTROL, 3.0 TO 7.5 HORSEPOWER SPINDLE DRIVE

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-M-80209B for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description covers vertical knee type milling machines having computer numerical control of x, y, and z (quill) movement with equipment and accessories as described herein.

1.2 Intended use. Milling machines covered by this commercial item description are intended for use in any metal cutting production shop where drilling, boring, tapping, reaming, and full (3-axis) contour milling operations are required.

2. CLASSIFICATION

2.1 Classification. The machine shall be of a vertical knee type having computer numerical control of x, y, and z (quill) axes. The machine shall be of the following classes, sizes, and grades. Class is determined by the spindle motor rating. Size is determined by the maximum table travel (X and Y axes) and quill travel (Z-axis). The class, size, and grade to be furnished shall be as specified (see 7.2).

Beneficial comments, recommendations, additions, deletions, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VCA, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

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2.2 Class and size.

Class A - 3.0 horse power spindle motor rating

Size	X-axis travel, maximum range	Y-axis travel, maximum range	Z-axis
1	24 in. (610 mm) to 25 in. (635 mm)	12 in. (305 mm) to 14 in. (356 mm)	5 in. (127 mm)
2	28 in. (711 mm) to 30 in. (762 mm)	12 in. (305 mm) to 15 in. (381 mm)	5 in. (127 mm)
3	30 in. (762 mm) to 32 in. (813 mm)	14 in. (356 mm) to 17 in. (432 mm)	5 in. (127 mm)

Class B - 5.0 horse power spindle motor rating

Size	X-axis travel, maximum range	Y-axis travel, maximum range	Z-axis
1	31 in. (787 mm) to 32 in. (813 mm)	14 in. (356 mm) to 15 in. (381 mm)	5 in. (127 mm)
2	36 in. (914 mm) to 38 in. (965 mm)	14 in. (356 mm) to 18 in. (457 mm)	5 in. (127 mm)

Class C - 7.5 horse power spindle motor rating

Size	X-axis travel, maximum range	Y-axis travel, maximum range	Z-axis
1	24 in. (610 mm) to 27 in. (686 mm)	14 in. (356 mm) to 15 in. (381 mm)	6 in. (152 mm)
2	30 in. (762 mm) to 32 in. (813 mm)	14 in. (356 mm) to 17 in. (432 mm)	6 in. (152 mm)

2.3 Grade. Grade A - High performance (see Table 1).
Grade B - Standard performance (see Table 1).

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be designed and manufactured in accordance with NFPA 79 and ANSI B11.8. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The machine configuration shall be consistent with current industry standards for a 3-axis, vertical knee type milling machine containing a column and base, knee, saddle, table, spindle, electric motor drive, and a computer numerical control (CNC) system. When specified (see 7.2), the machine shall be of a specific configuration. The control system shall be a continuous path - closed looped system providing full contouring capabilities on the machine's three linear axes. The term "full contouring", used herein, shall be defined as a closed loop control system having simultaneous control of position, direction, and rate of movement of the machine's three linear axes in linear, parabolic, helical, and circular interpolation. The machine shall provide a means for overload protection. The machine shall automatically stop spindle rotation and the motion of all slides upon an overload condition or spindle stall. The machine shall have a means to permit leveling and mounting to the floor or foundation. The axes of the machine shall be identified in accordance with EIA-267.

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3.2 Measurement systems. Either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 7.2).

3.2.1 Measuring and indicating device. Either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.3 Table. The table size shall be the manufacturer's standard table size and shall have three T-slots, 0.625 inches (15.9 mm) wide and 2.25 inces (57 mm) or 2.5 inches (63.5 mm) from center to center. The maximum rapid traverse speed shall not be less than 175 in/min (44.5 m/min) and the maximum feedrate shall not be less than 100 in/min (25.4 m/min). Unless otherwise specified (see 7.2), the table travel (X and Y axes) shall be as specified in section 2.2 for the class and size indicated. When specified (see 7.2), any combination of the following items shall be provided.

a. Rotary table as specified (see 7.2).
b. Index table as specified (see 7.2).
c. Work light.

3.4 Spindle. Unless otherwise specified (see 7.2), the minimum spindle speed range shall be from 100 to 3500 revolution per minute in both clockwise and counterclockwise direction. The spindle taper size shall be in accordance with ANSI B5.18. Unless otherwise specified (see 7.2), the quill travel (Z-axis) shall be as specified in section 2.2 for the class and size indicated.

3.5 Knee. Unless otherwise specified (see 7.2), the knee shall have a range of travel of not less than 14 inches (356 mm). The knee shall be provided with a means for preventing dust and chips from entering the ways.

3.6 Lubrication. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. All oil holes, grease fittings, and filler caps shall be accessible. When provided, recirculating lubrication system shall have a replaceable in-line filter capable of removing particles 10 microns in size. A means for determining lubricant level in the reservoir(s) shall be by sight gauge, warning light, or a similar device.

3.7 Coolant System. Unless otherwise specified (see 7.2), the machine shall have a flood type coolant system. A means shall be provided to direct and control the amount of coolant. A base or floor-mounted splash tray shall be provided to catch coolant overflow from the chip and coolant channels around the working surface of the table.

3.8 Computer numerical control (CNC) system. The machine shall have a fully automated CNC system. The CNC system shall operate as a continuous-path closed loop system providing full contouring capabilities. The processing speed and temporary memory storage shall be of sufficient

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capacity to prevent delaying of the next incoming command or interrupting the machine's functions. The CNC system shall provide on-line diagnostics for detecting programming errors and problems with the control system. The CNC system shall be furnished with a postprocessor for converting specified part program language (see 7.2) to the form and format required for input to the CNC system. The CNC system shall be in accordance with all safety guidelines and procedures as specified herein.

3.8.1 Hardware features. Unless otherwise specified (see 7.2), the CNC system shall have not less than the following hardware features:

a. Operator control panel.
b. High resolution 9 inch monochrome display.
c. 3.5 inch, 1.44 Mbyte floppy disk drive.
d. 100 Mbyte hard-drive.
e. EIA RS-232C serial port.

3.8.1.1 Operator control panel. Unless otherwise specified (see 7.2), the operator control panel shall have not less than the following control features:

a. Emergency stop
b. Feed hold
c. Cycle start
d. Feed rate override
e. Spindle speed override
f. Axis jog (continuous and intermittent)
g. Single block read
h. Dry run
i. Mode select

3.8.1.2 Data display unit. Unless otherwise specified (see 7.2),the data display unit shall display not less than the following information:

a. Block sequence number
b. Active block of stored data
c. Actual position of each axis
d. Feedrate and override condition
e. control/system error messages
f. Graphics simulation (4 views) - XY, XZ, YZ, and isometric

3.8.2 Operational modes.

3.8.2.1 Automatic mode. The automatic mode shall set the CNC system for full automatic "hands-off" program operation.

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3.8.2.2 Manual data input (MDI) mode. The MDI mode shall permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions in accordance with EIA RS-274. The data shall be immediately displayed on the data display screen to permit data verification. The MDI mode shall have full editing capabilities. A means shall be provided for saving and recalling a part program developed in the MDI mode.

3.8.2.3 Conversational mode. The conversational mode shall provide the operator the ability to develop part programs by following computer prompts. A means shall be provided for saving and recalling a part program developed in the conversational mode.

3.8.3 Axis and tool management. Unless otherwise specified (see 7.2), the CNC system shall have not less than the following axis and tool management features:

a. Tool length offsets
b. Cutter compensation
c. Fixture compensation
d. Reversal error compensation
e. Lead error compensation

3.8.4 Program and control features. Unless otherwise specified (see 7.2), the CNC system shall have not less than the following program and control features:

a. Spindle - off/clockwise/counter clockwise.
b. Coolant - on/off.
c. Reference zero.
d. Sequence number search.
e. Switchable inch/metric conversion.
f. Cartesian/polar coordinates.
g. Scale factor.
h. Mirror image.
i. Absolute/incremental axis positioning.
j. Interpolation - linear, circular, and helical.
k. Canned cycles - drilling, boring, tapping, and bolt hole.
l. Part rotation.
m. Dwell.

3.8.5 Variable data block format. The preparatory “G codes” and miscellaneous functions “M codes” shall be in accordance with EIA RS-274.

3.8.6 Program safe limits. The control system shall have the capability to place boundaries in a program beyond which the tool will not travel.

3.8.7 Battery back-up. When specified (see 7.2), the CNC system shall have a battery back-up with a storage capacity as specified (see 7.2).

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3.9 Electrical requirements. The milling machine's electrical system shall conform ANSI/NFPA-79. The machine shall operate from a single 230/460 volt, 3 phase, 60 hertz (Hz) circuit. Unless otherwise specified (see 7.2), the machine shall be initially wired for 230 volt operation. All motors shall be energy efficient and shall conform to NEMA MG-1. The CNC system shall operate on a 120 volt, single phase,60 Hz circuit.

3.10 Performance. Unless otherwise specified (see 7.2), the milling machine shall meet the performance requirements specified in Table I when tested in accordance with section 5.3 through 5.3.4 herein.

TABLE I - Performance accuracy

Test Name	Parameter	Performance Category	
		Grade A	Grade B
Linear Displacement Accuracy	X, Y, and Z directions	0.0005 in/40 in. (0.0125 mm/m)	0.0020 in/40 in. (0.0500 mm/m)
Bi-directional Repeatability	X, Y, and Z directions	±0.0002 in. (±0.0050 mm)	±0.0005 in. (±0.0125 mm)
Volumetric Performance	Diagonal Displacement	0.0016 in/40 in (0.0400 mm/m) for each axis	0.0040 in/40 in (0.1000 mm/m) for each axis
Contouring Performance	X-Y Plane, 360° X-Z Plane, 190° Y-Z Plane, 190°	0.0008 in. (0.0200 mm) up to 80% of the maximum speed	0.0020 in. (0.0500 mm) up to 80% of the maximum speed
Periodic Angular Positioning Accuracy		5.0 arc-sec	5.0 arc-sec
Bi-directional Angular Positioning Repeatability		10.0 arc-sec	10.0 arc-sec
Spindle Dynamics	Radial error (Average)	.0003 in. (0.0076 mm)	.0003 in. (0.0076 mm)
	Radial error (Asynchronous)	.0005 in. (0.0127 mm)	.0005 in. (0.0127 mm)

Note: The machine's performance shall not be greater than the quantities specified for the performance category indicated. If the axis travel is less than 40 inches, the tolerance shall be decreased proportionately.

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3.10 Standard equipment. The manufacturer shall furnish all equipment considered standard for his machines.

3.11 Optional equipment. Optional equipment shall be furnished as specified (see 7.2).

3.12 Safety and health requirements. The manufacturer shall ensure the machine and all equipment and accessories used on the machine shall be in compliance with ANSI B11.8, OSHA 1910.212, and OSHA 1910.219. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.13 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2), the nameplate shall contain the following information:

Nomenclature
Manufacturer's name
Machine model designation
Power input (volts, total amps, phase, frequency)
Short circuit/over-current rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.14 Lubrication plate or chart. When specified (see 7.2), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

Points of lubricant application
Servicing interval
Type of lubricant(s) with SAE number or lubricant identifier

4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirement, and promotes economically advantageous life cycle costs.

4.2 Environmental protection. The machine shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health and shall be met during the manufacturing, service, transportation, storage, and operation of the machine.

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5.0 QUALITY ASSURANCE PROVISIONS

5.1 Contractor Certification. The contractor shall certify and maintain evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Examination. The machine shall be examined to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2), the machine shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Test conditions. The machine shall be tested in accordance with the environmental conditions as specified in section 3 of ASME B5.54.

5.3.2 Operational test. The machine and its CNC system shall be operated in accordance with the manufacturer's standard operating procedures for warm-up and run-off checks. The machine shall be operated at no load for not less than 3 hours. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The performance tests shall be conducted in accordance with Appendix B of ASME B5.54. If a rotary table is included with the milling machine, the periodic angular positioning accuracy and the bi-directional angular positioning repeatability tests shall be conducted in accordance with ASME B5.54. The test results will be compared to the performance levels listed in Table I.

5.4 Acceptance. Unless otherwise specified (see 7.2), preliminary acceptance testing shall be conducted at the origin (manufacturer's site) and final acceptance testing shall be conducted at the destination (government site). The results of the performance tests will be compared to the performance criteria listed in Table I. The machine shall meet or exceed the criteria listed in Table I for the performance category selected. If the contractor's test environment has been met and the machine does not meet the thresholds prescribed herein, the contractor shall either make repairs to bring the machine into compliance or replace it with a new machine.

6. PACKAGING

6.1 Packaging requirements. Unless otherwise specified (see 7.2), the machine will be packaged in accordance with the manufacturer's standard practices.

7. NOTES

7.1 Referenced documents.

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7.1.1 Government documents. Copies of Military and Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.
7.1.2 Industry/commercial documents. Copies of industry documents/standards referenced in this commercial item description may be obtained from the following addresses:

a. American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.
b. American Society of Mechanical Engineers (ASME), 22 Law Drive, Fairfield, NJ 07007.
c. National Fire Protection Association (NFPA), Batterymarch Park, Quincy, MA 02269.
d. National Electrical Manufacturer's Association (NEMA), 2101 L Street, Washington D.C. 20037.
e. Occupational Safety and Health Association (OSHA), see section 7.1.1 or from Commerce Clearing House, Inc., 4025 West Peterson Avenue, Chicago, IL 60646.
f. EIA standards are available from Global Engineering Documents, 15 Inverness Way East, Englewood, CO 80112-5776.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Class, size, and grade of machine required (see 2.1).
c. Machine configuration (fixed head and column, swivel head, ram, turret), if required (see 3.1).
d. Specific measurement system, if required (see 3.2).
e. Specific measurement system for indicating devices, if required (see 3.2.1).
f. Dual graduation of indicating devices, if required (see 3.2.1).
g. Table travel (X and Y axes), if different (see 3.3).
h. Rotary table, if required (see 3.3).
i. Index table, if required (see 3.3)
j. Work light, if required (see 3.3).
k. Spindle speed, if different (see 3.4).
l. Quill travel (Z-axis) for Class A, B, and C machine, if different (see 3.4).
m. Knee travel, if different (see 3.5)
n. Coolant system, if different (see 3.5).
o. Part program language used, specify (see 3.8).
p. Hardware features, if different (see 3.7.1).
q. Operator control panel, if different (see 3.8.1.1).
r. Data display unit, if different (see 3.8.1.2).
s. Axis and tool management features, if different (see 3.8.3).
t. Program and control features, if different (see 3.8.4).
u. Battery back-up and storage capacity, if required (see 3.8.7).
v. Electrical requirements, if different (see 3.9).
w. Performance requirements, if different (see 3.10).
x. Optional equipment, if required (see 3.11).
y. Nameplate, if different (see 3.13).
z. Lubrication chart, if required (see 3.14).
aa. Performance testing, if different (see 5.3).

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x. Optional equipment, if required (see 3.11).
y. Nameplate, if different (see 3.13).
z. Lubrication chart, if required (see 3.14).
aa. Performance testing, if different (see 5.3).
bb. Acceptance criteria, if different (see 5.4).
cc. Packaging, if different (see 6.1).

7.3 Part Identification Number (PIN) The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

AAXXX-XX

Grade: 1- Grade A
2- Grade B

Class/size: 1- Class A, size 1
2- Class A, size 2
3- Class A, size 3
4- Class B, size 1
5- Class B, size 2
6- Class C, size 1
7- Class C, size 2

MILITARY INTERESTS:

Custodians
Air Force - 99
Navy - SH

Rewiew Activities
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

PREPARING ACTIVITY:
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